Applied Catalysis A, General 602 (2020) 117607

Contents lists available at ScienceDirect

Applied Catalysis A, General

journal homepage: www.elsevier.com/locate/apcata

# Carbon-Carbon bond formation during Fe catalyzed Fischer-Tropsch synthesis

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ARTICLE INFO

*Keywords:*
Fischer-Tropsch synthesis
Inverse isotope effect
Deuterium enrichment
Alkylidene mechanism
Deuterium tracer

ABSTRACT

An inverse isotope effect and deuterium enrichments in hydrocarbons were observed in Fe catalyzed Fischer-Tropsch (FT) reactions. These results are explained by an Alkylidene mechanism by which the monomer of the polymerization-like reaction is M≡CH and the growing chain is RCH = M. The formation of C−C bond is through the reaction of M≡CH and RCH = M, producing a new growing chain $RCH_2CH$ = M. During the process of C−C bond formation, the $sp^2$ carbon in RCH = M is changed to $sp^3$ in newly formed growing chain $RCH_2CH$ = M, which results in an inverse isotope effect. The rate of reaction of RCD = M with M≡CH/M≡CD is fast than that of RCH = M with M≡CH/M≡CD. The calculations based on alkylidene mechanism showed that the deuterium is enriched in hydrocarbons and that the deuterium enrichment is a function of carbon number.

## 1. Introduction

Fischer-Tropsch (FT) synthesis is a complex polymerization-like reaction in which two simplest reagents CO and $H_2$ are used as the starting reagents, producing a mixture that includes alkanes, alkenes and oxygenates ranged from $C_1$ to more than $C_{60}$. The experimental facts observed before 1970s showed that the product distribution of the FT reactions follows Anderson-Schulz-Flory (ASF) equation [1]. However, with the development of Gas Chromatograph and other analytical techniques that allow researchers to analyze the FT products up to $C_{60}$, several research groups reported the product distribution deviated from the ASF kinetics in 1980s [2–4]. To explain the positive or negative deviations from ASF distribution and the unusual paraffin to olefin ration (P/O), several theories were proposed [4,5]. In 2000s, Dr. Davis' group reported that deuterated octane does not undergo H/D exchange under FT reaction conditions and only 5 deuterium atoms in deuterated octene undergo H/D exchange [6]. This finding opened a door to use deuterium gas as the tracer to study the mechanism for FTS. By utilizing the deuterium tracer techniques, Dr. Davis' research group showed that the deviations from the ASF distribution is a result of product accumulation [7–9]; the increasing of P/O ratio with the carbon number is much smaller by $H_2/D_2$ switching experiment than reported [10]. Thus, the ASF equation can still be used to predict the true distribution of FT products.

Since there is no H/D exchange in alkane products of FT reactions, the deuterium tracer techniques becomes a powerful tool in studying the mechanism of FT reactions. By utilizing the $H_2/D_2$ switching technique, we confirmed the inverse isotope effect reported by several groups in Co catalyzed FT reactions [11]. We also observed that there exists the deuterium enrichment in hydrocarbons produced by Co catalyzed FT reactions and the deuterium enrichment is a function of carbon number [11].

To account for the inverse isotope effect and deuterium enrichment in hydrocarbons produced by Co catalyzed FT reactions, we proposed an Alkylidene mechanism for FTS, in which R−CH = M is the growing chain and M≡CH is the monomer [11,12]. This mechanism was also used to explain the formation of 2-alkenes as the secondary products of FTS [13] and the formation of branched hydrocarbons produced during FTS [14].

According to Alkylidene mechanism, the $sp^2$ hybridized carbon in growing chain RCH = M is changed to $sp^3$ hybridized carbon in new growing chain $RCH_2CH$ = M after one more propagation step. It is the change in hybridization that produced an inverse isotope effect if hydrogen atom in RCH = M is replaced by deuterium atom in RCD = M when syngas $CO/H_2$ is switched to $CO/D_2$. When syngas $CO/H_2/D_2$ is used, the competition of RCH = M and RCD = M in each propagation step will result in deuterium enrichment in hydrocarbons. Thus, both the inverse isotope effects and deuterium enrichment during FTS are originated from the C−C bond formation steps.

It has been reported that there is an inverse isotope effect during Fe

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https://doi.org/10.1016/j.apcata.2020.117607
Received 1 January 2020; Received in revised form 20 April 2020; Accepted 28 April 2020
Available online 15 May 2020
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catalyzed FT reactions [15,16]. Whether there exists the deuterium enrichment and whether the deuterium enrichment is a function of carbon number during Fe catalyzed FT reactions are still not reported. In this study, we report the results of the deuterium tracer experiments and provide the explanation on how the C—C bonds are formed in Fe catalyzed FT reactions.

## 2. Experimental

### 2.1. Preparation of catalysts

Synthesis of Fe/Si/K catalysts were followed the procedure from the literature [17] with minimal modification. The following is a typical catalyst synthesis procedure: 40 g of $Fe(NO)_3 \cdot 9H_2O$ was dissolved in 40 mL of distilled water while stirring. Then, 1 g of tetraethylorthosilicate ($Si(OC_2H_5)_4$) was added into the $Fe(NO)_3$ solution. A layer of oil was observed after the addition of tetraethylorthosilicate. The mixture was stirred vigorously for 4−6 h till the tetraethylorthosilicate totally hydrolyzed. The solution was added to another flask by pipette with the addition of $NH_4OH$ at the same time to keep the formed slurry at pH 9 while stirring. The result slurry was then vacuum filtered. The filter cake was washed with 20 mL of distilled water three times. The filter cake was oven dried for 24 h at 110 °C. The filter cake was then mashed into fine powder. Fifty (50) mg of $K_2CO_3$ was dissolved in 5 mL of distilled water. The $K_2CO_3$ solution was then added into the previous powder with stirring. The mixture was oven dried for 24 h at 110 °C. The crude catalyst was then calcinated at 350 °C for 4 h to burn the organic matters.

The Fe/Si catalysts were synthesized with the similar method but without the addition of $K_2CO_3$ solution. The crude catalysts were sent to calcination for 4 h directly after 24 h oven dry at 110 °C. The Fe/Si (100 : 4.6) catalyst was obtained.

### 2.2. Fischer-Tropsch synthesis

The Fischer-Tropsch synthesis procedures by $H_2/D_2$ switching method and $H_2/D_2$ competition method are the same as described previously [11]. In a typical run, 1.5 g of iron catalyst (Fe:Si:K = 100:4.6:1.4) is used. A mixture of $H_2$ (60%), CO (30%) and $N_2$ (10%) was used as the syngas. The reactions were conducted at 1.3 Mpa and 270 °C. After the CO conversion reaches the desired conversion level, the liquid sample was collected every 4−8 h for 120 h. The syngas then was switched to a mixture of $D_2$ (60%), CO (30%) and $N_2$ (10%). After 5 liquid samples (total 120 h) were taken, the syngas was switched back to $CO/H_2/N_2$ again for 60 h. For the $H_2/D_2$ competition experiments, a mixture of CO (30%), $H_2$ (30%), $D_2$ (30%) and $N_2$ (10%) was used, and the liquid samples from “hot trap” and “cold trap” were collected every 4−8 h. The competition experiments were usually conducted for 90−96 h.

### 2.3. Analysis of products

The gas samples were collected every 2 h and analyzed using Agilent 3000A Micro GC. The CO conversions were determined.

The liquid samples from $H_2/CO/N_2$ runs (a mixture of samples from “hot trap” and “cold trap”) were collected every 4−8 h and analyzed by a Thermo Focus GC with a 60 miter SPB-5 column and the identities of products were confirmed by HP GC/MS with a 60 miter SPB-5 column. The product distribution can be obtained. The calculation of the relative amount of isotopemers of competition experiments is based on total area of molecular ion of each isotopemer after making $^{13}C$ correction.

**Table 1**
CO conversions during $H_2/D_2$ switching experiments (Temperature: 270 °C; Pressure: 188 psi).

| Run | Catalyst* | GHSV | CO conversion (%) H$_2$/CO | D$_2$/CO | H$_2$/CO | $[CO_{conv}]^H/[CO_{conv}]^D$ |
|---|---|---|---|---|---|---|
| 1 | Fe/Si/K | 1.748 | 59.5±1.9 | 70.8±1.7 | 61.6±1.7 | 0.86±0.02 |
| 2 | Fe/Si/K | 2.257 | 11.8±1.7 | 27.5±2.4 | 17.0±0.6 | 0.52±0.13 |
| 3 | Fe/Si/K | 1.403 | 78.3±0.9 | 84.8±0.6 | 77.5±0.6 | 0.92±0.01 |
| 4 | Fe/Si | 1.765 | 8.2±0.9 | 18.8±2.2 | N/A | 0.44 |

*Catalysts used in Run 1 and Run 3 were from the same batch of catalyst; catalyst in Run 2 was from a different batch of catalyst.

## 3. Results and discussions

### 3.1. Inverse isotope effect during Fe catalyzed fischer-tropsch synthesis

The results from $H_2/D_2$ switching experiment are given in Table 1. In Run 1, the CO conversion was 59.5% during about 120 h on stream when $CO/H_2$ was used; then the syngas was switched to $CO/D_2$ for another 120 h, the CO conversion increased to 70.8%; after switching back to $CO/H_2$ for about 60 h, the CO conversion decreased to 61.6%, which is about the same as before switching to $CO/D_2$, indicating the catalyst is stable during about 300 h on stream. The ratio of $[CO_{conv}]^H/[CO_{conv}]^D$ is 0.86. Since the $CO/H_2$ and $CO/D_2$ runs were under the same reaction conditions, the ratios of CO conversions should be related to the inverse isotope effect. In Run 3, the catalyst used is the same as in Run 1. When GHSV decreased to 1.403 from 1.748 in run 1, the CO conversion is 78.3% when $CO/H_2$ was used as the syngas; after switching to syngas $CO/D_2$, the CO conversion is 84.8%; after switching back to syngas $CO/H_2$, the CO conversion is 77.5%. In Run 2, the catalyst used is a different batch catalyst from Run 1 and Run 3, much less active than the catalyst used in Run 1 and Run 3. With GHSV of 2.257, the CO conversion in $CO/H_2$ run is 11.8%. After switching to $CO/D_2$ run, the CO conversion increased to 27.5%. The CO conversion decreased to 17.0% after switching back to $CO/H_2$. For Run 4, after switching back to $CO/H_2$ from $CO/D_2$, the conversion is so low that the measured CO conversions was not accurate, but it also indicated an inverse isotope effect. As shown in Table 1, the ratios of $[CO_{conv}]^H/[CO_{conv}]^D$ are in a range of 0.44-0.92. Similar results were also reported by other research group using similar method [15].

During FT reaction, the CO is converted to $CO_2$, $CH_4$ and $C_1^+$ (products with 2 carbon atoms or more) products. The formation of $CO_2$ is an oxidation process, the formation of $CH_4$ is a reduction process involving C–H bond formation, and the formation of $C_1^+$ involves the C—C bond formation. The question whether the inverse isotope effect that resulted in the ratios of $[CO_{conv}]^H/[CO_{conv}]^D$ less than 1 is a result of the formation of $CO_2$, or $CH_4$, or $C_1^+$ can be answered from the data in Table 2. Table 2 is the product distribution in each switching run. The inverse isotope effect cannot be the result of $CO_2$ formation since

**Table 2**
Product Distribution in $H_2/D_2$ switching experiments.

| Run* | Syngas | $CO_{conv}$% | $CO_2$% | $CH_4$% | $C1^+$% | $[HC]^H/[HC]^D$ | $[C1^+]^H/[C1^+]^D$ |
|---|---|---|---|---|---|---|---|
| 1 | $CO/H_2$ | 60.41 | 36.44 | 3.67 | 20.30 | 0.582 | 0.544 |
| | $CO/D_2$ | 70.82 | 29.66 | 3.87 | 37.29 | | |
| 2 | $CO/H_2$ | 14.59 | 3.66 | 1.65 | 9.28 | 0.461 | 0.435 |
| | $CO/D_2$ | 27.08 | 3.39 | 2.34 | 21.35 | | |
| 3 | $CO/H_2$ | 77.77 | 10.98 | 6.74 | 60.05 | 0.887 | 0.887 |
| | $CO/D_2$ | 84.82 | 9.54 | 7.56 | 67.72 | | |
| 4 | $CO/H_2$ | 8.63 | 2.41 | 2.12 | 4.10 | 0.360 | 0.288 |
| | $CO/D_2$ | 19.60 | 2.31 | 3.07 | 14.22 | | |

*the catalyst and the reaction conditions are the same as the ones listed in Table 1.

more $CO_2$ was produced when the syngas is $CO/H_2$ than the syngas is $CO/D_2$ in each switching run, which should result in a normal isotope effect. The formation of $CH_4$ can produce an inverse isotope since amount of $CH_4$ increased when switching to $CO/D_2$. In run 1, after switching to $CO/D_2$, the amount of $CH_4$ increased from 3.67% to 3.87%. However, the CO conversion increased from 60.41% to 70.82%, and the increases in $CH_4$ formation cannot account for the increase in CO conversion. The big increase is from the formation of $C_1^+$ products after switching to $CO/D_2$ in which the amount of $C_1^+$ increased to 37.29% from 20.30%. The Run 2–4 show the similar results. These results indicate that the increased CO consumption after switching to $CO/D_2$ is mainly used to produce $C_1^+$ products. Thus, the ratio of $[C_{1+}]^H/[C_1^+]^D$ in each switching run can be used to relate the inverse isotope effect of the FT reaction. The ratios of $[C_1^+]^H/[C_1^+]^D$ in each rum are presented in Table 2. In addition, the ratios of hydrocarbon formation ($C_1^+$ plus $CH_4$) $[HC]^H/[CH]^D$ are also presented in the table. As expected, the ratio $[HC]^H/[CH]^D$ in each switching run is not significantly alter the value of $[C_1^+]^H/[C_1^+]^D$. These results indicate that there is an inverse isotope effect in the formation of $C_1^+$ products during Fe catalyzed FT reactions.

One of explanations for the inverse isotope effect is that the FT reaction is through H-assisted CO dissociation pathway [15], and the deuterium accelerated the CO dissociation. An alternative explanation is that the inverse isotope effect is the result of C—C bond formation during propagations of the FT reaction. This explanation is based on the fact that the increased CO consumption is mainly used to produce the $C_1^+$ FT products.

The ratios of CO conversion $[CO_{conv}]^H/[CO_{conv}]^D$ or the ratios of hydrocarbon formation $[HC]^H/[HC]^D$ or $[C_1^+]^H/[C_1^+]^D$ are related to the inverse isotope effect of the FT reaction, but those are not the actual value of the inverse isotope effect. As described previously [11], the values of $[CO_{conv}]^H/[CO_{conv}]^D$, $[HC]^H/[HC]^D$ and $[C1^+]^H/[C1^+]^D$ are the result of an accumulated value of the true inverse isotope effect originated from C—C bond formation in each propagation step of the FT reaction.

According to ASF kinetics, the rate of formation of a hydrocarbon $C_n$ ($r_n$) = $r_1\alpha^{n-1}$, where n is the carbon number. Since the formation of methane does not involve the C—C bond formation, and the C—C formation is related to the $C_1$ growing chain formation, the ASF equation can be modified as $r_n = C_1\alpha^{n-1}$. The rate of formation of $r_n$ for $CO/H_2$ run becomes Eq. (1),

$$r_n^H = C_1^H\alpha_H^{n-1} \quad (1)$$

For $CO/D_2$ run, we have Eq. (2),

$$r_n^D = C_1^D\alpha_D^{n-1} \quad (2)$$

If we assume that the rates of formation of growing chain $C_1$ for $CO/H_2$ run and $CO/D_2$ run are the same, $C_1^H = C_1^D$, we have Eq. (3),

$$\frac{r_n^H}{r_n^D} = \frac{C_1^H\alpha_H^{n-1}}{C_1^D\alpha_D^{n-1}} = \left(\frac{\alpha_H}{\alpha_D}\right)^{n-1} \quad (3)$$

If the $C_k$ hydrocarbon is the one with average molecular mass of the FT reaction, the rate of formation of $C_k^H$ equals to the CO conversion by assuming that all of converted CO is used to produce $C_k$ hydrocarbon. Thus, the Eq. (3) becomes Eqs. (4) and (5), where carbon number k can be estimated from the product distribution.

$$\frac{[CO_{conv.}^H]}{[CO_{conv.}^D]} = \left(\frac{\alpha_H}{\alpha_D}\right)^{k-1} \quad (4)$$

$$\frac{\alpha_H}{\alpha_D} = \sqrt[k-1]{\frac{[CO_{conv.}^H]}{[CO_{conv.}^D]}} \quad (5)$$

During FT reactions, the consumed CO molecules are not only used to produce hydrocarbons, significant amount of these consumed CO are converted to $CO_2$. If only the hydrocarbons are considered, the Eq. (5) becomes Eq. (6), where [HC] is the amount of FT products by excluding the amount of $CO_2$.

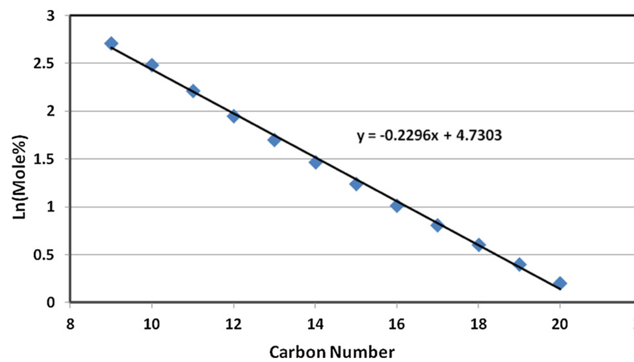



**Fig. 1.** Product distribution of the FT reaction catalyzed by Fe/Si/K.

$$\frac{\alpha_H}{\alpha_D} = \sqrt[k-1]{\frac{[HC]^H}{[HC]^D}} \quad (6)$$

Since the formation of methane does not involve C—C bond formation, it is reasonable to exclude the amount of $CH_4$ from the total amount of hydrocarbon products when considering the inverse isotope effect. Thus, the Eq. (6) can be changed to Eq. (7), where the $[C_1^+]$ is amount of FT products with 2 and more carbon atoms, and k' is the carbon number of the compound $C_{k'}$ with average molecular mass of the FT products with 2 and more carbon atoms.

$$\frac{\alpha_H}{\alpha_D} = \sqrt[k'-1]{\frac{[C_1^+]^H}{[C_1^+]^D}} \quad (7)$$

The k and k' value can be estimated by using the product ASF distribution. For example, Fig.1 is the product ASF distribution for Fe/Si/K catalyzed FT reaction during $CO/H_2$ run. Based on the result of Fig. 1, the α value for this reaction is 0.80. If we assume that the FTS can produce only alkanes from $C_1$ to $C_{60}$, the number of moles of alkanes in every carbon number can be calculated based on $C_n = r_1\alpha^{n-1}$. Then the mole fraction of each alkane can be calculated and the mass of each alkane can be obtained by multiplying the molar mass of the alkane by the mole fraction. The total mass of the alkanes from $C_1$ to $C_{60}$ will be the average molecular mass of the alkanes produced by the FT reaction. The k value can be obtained by assuming that the molecular formula is $C_kH_{2k+2}$. The average molecular mass of $C_1^+$ can also be obtained by subtracting the amount of methane and the k' value can be obtained by assuming that the molecular formula is $C_{k'}H_{2k'+2}$.

For Fe/Si/K catalysts, the α value is 0.80 in Run 1 and Run 3; for the catalyst from a different batch of Fe/Si/K catalyst used in Run 2, the α value is also 0.80. The average molecular mass is 71.9 with a molecular formula $C_kH_{2k+2}$, where the k is 4.99. After the amount of $CH_4$ is excluded, the average molecular mass for $C_1^+$ is 68.7 with molecular formula $C_{k'}H_{2k'+2}$, where the k' is 4.77. For Fe/Si catalyst, the α value is 0.74 and the average molecular mass is 55.8 and the average molecular mass for $C_1^+$ is 51.7, and the corresponding k and k' values are 3.85 and 3.55, respectively. By using the ratios of CO conversion, hydrocarbon formation and $C_1^+$ products formation during the $H_2/D_2$ switching runs in Tables 1 and 2, the inverse isotope effect of each run can be calculated based on Eq. 5–7. These results are summarized in Table 3.

In Table 3, the $CO_{conv}$% is the CO conversion during $CO/H_2$ run; the α value is based on the product distribution during $CO/H_2$ run; the average molecular mass (A.M.M) is calculated by assuming all of the FT products are alkanes, olefins and oxygenates were not considered and the value in parentheses is the average molecular mass of alkanes with excluding the amount of $CH_4$; and the k value is based on the formation of alkanes; the k' values were obtained by excluding the amount of $CH_4$.

**Table 3**
Inverse isotope effects based on CO conversion and hydrocarbon formation.

| Run* | $CO_{con}$% | α | A.M.M** | K(k')# | $k-\sqrt[1]{\frac{[CO^{H}_{conv}]}{[CO^{D}_{conv}]}}$ | $k-\sqrt[1]{\frac{[HC]^{H}}{[HC]^{D}}}$ | $k'-\sqrt[1]{\frac{[C_1^{+}]^{H}}{[C_1^{+}]^{D}}}$ |
|---|---|---|---|---|---|---|---|
| 1 | 60 | 0.80±0.02 | 71.9 (68.7) | 4.99 (4.77) | 0.963 | 0.873 | 0.851 |
| 2 | 15 | 0.80±0.02 | 71.9 (68.7) | 4.99 (4.77) | 0.849 | 0.823 | 0.802 |
| 3 | 77 | 0.80±0.01 | 71.9 (68.7) | 4.99 (4.77) | 0.929 | 0.970 | 0.968 |
| 4 | 8 | 0.74±0.04 | 55.8 (51.7) | 3.85 (3.55) | 0.749 | 0.698 | 0.614 |

*the catalyst and reaction conditions are the same as the ones listed in Table 1. **A.M.M: Average Molecular Mass; the values in parentheses is the average molecular mass of $C_1^{+}$ products. #the value in parentheses is the number of carbon for $C_1^{+}$ alkanes.

In deriving Eq. (5), it is assumed that the consumed CO is converted to hydrocarbons only. This assumption is not consistent with the experimental results, where the amount of $CO_2$ in all of the runs are more than 10% of all products. Whether there is an inverse isotope effect or normal isotope or no isotope effect in the formation of $CO_2$, it will have significant impact on the ratio of CO conversion when the amount of $CO_2$ is included in calculations. Therefore, the inverse isotope effect based on CO conversion will significantly deviate from the true value. The inverse isotope effects in $C_1^{+}$ products calculated based on Eq. (7) are in the range from 0.614 to 0.968. For Run 1, the inverse isotopes is 0.851, and for run 2, the value is 0.802, which is very close to the value in Run 1 even through the catalysts are from different batches and have different activity. However, the isotope effect obtained from Run 3 is 0.968, which is larger (smaller inverse isotope effect) than Run1 in spite of the same catalyst used in these two runs. In run 3, the CO conversion is almost 85% after switching syngas to $CO/D_2$, which may be the maximum of the CO that can be converted under the reaction conditions. It is understandable that if the conversion is very close to the maximum of the CO conversion, the measured inverse isotope effect based on conversion becomes smaller. For Run 4, the Fe catalysts used is different from the catalysts used in run 1–3, at lower conversion level, the inverse isotope effect is o.614. When the amount of $CH_4$ is included, the inverse isotope effect for the formation of hydrocarbons in each run is slightly smaller (slightly larger value) than the value for $C_1^{+}$, agreeing with our argument that small amount of $CH_4$ will not significantly alter the results.

Even though the values of isotope effects calculated are scattered, they are roughly in a range of 0.7 to 0.95, which are consistent with the value of an inverse isotope effect of a reaction in which the $sp^2$ C–H in the reagent is changed to $sp^3$ C–H in the product and have an inverse isotope effect that can be calculated by Eq. (8),

$$\frac{k_H}{k_D} = \exp[-\frac{0.1865}{T}(\Delta\nu)] \tag{8}$$

where the maximum value of an inverse isotope effect is when $\Delta\nu$ = (1350-800). If the $\Delta\nu$ becomes smaller, the $k_H/k_D$ will be closer to 1.

These results suggest that the values of $[CO_{conv}]^H/[CO_{conv}]^D$ or $[HC]^H/[HC]^D$ or $[C_1^{+}]^H/[C_1^{+}]^D$ are not the value of an inverse isotope effect. Rather, they are the accumulated effect of the inverse isotope effect during FT reaction and there is an inverse isotope effect in each step of the propagations.

It is worth to point out that the more accurate way to determine the inverse isotope effect is to determine the $k_H/k_D$ for some FT products. Because so many factors affect the relative ratio of the CO conversion, the $H_2/D_2$ switching method utilized in this study can only determine whether there is an inverse isotope effect and what is the roughly range of the effect. Since the inverse isotope effects obtained in this study is in the range of the well-established inverse isotope effect for organic reactions, it may not be necessary to invoke the H-assisted CO dissociation pathway to explain the inverse isotope effect observed during FT reactions.

The inverse isotope effect in FT reactions can be explained by the alkylidene mechanism as shown in Scheme 1, where the growing chain is R−CH = M and the monomer is M≡CH. As indicated in Scheme 1, the inverse isotope effect (IIE) originated in each step of the propagations, and the value equals to $f_H/f_D$ by assuming there is no preference either reacting with MH or MD and $f_h/f_d$ = $\alpha_H/\alpha_D$.

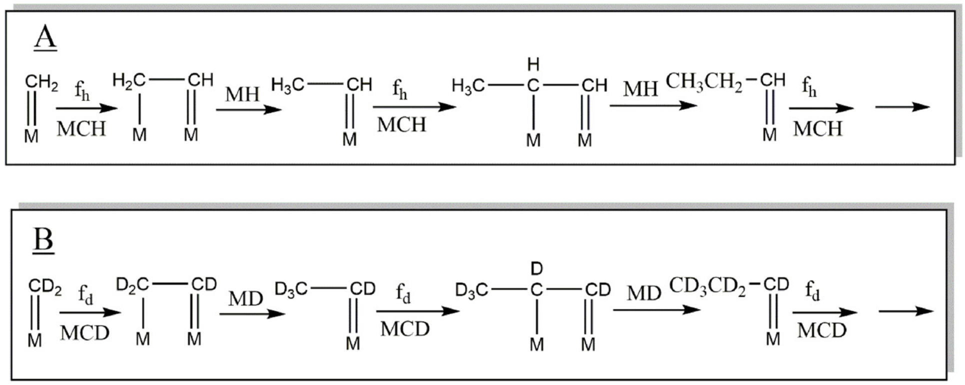
### 3.2. Deuterium Enrichment in hydrocarbons and the Deuterium Enrichment as a Function of Carbon number

The competitive experiments were conducted by using equal molar of $CO/H_2/D_2$ (1:1:1) as the syngas. If there is no isotope effect and the behavior of $D_2$ is the same as the $H_2$, the H/D ratio in each hydrocarbons produced by the competition experiemnt should be 1. If the inverse isotope effect occurs prior to the C−C bond formation steps, the H-containing growing chain and the D-containing growing chain should have the same opportunity to compete for the dissociated CO species wheter it is H-assisted or D-assisted. The H/D ratio in each hydrocarbons should also be 1 or less than 1 but a constant for all hydrocarbon if more deuterium containing species are produced prior to propagations. If the inverse isotope effect occurs in the termination step, the deuterium will be enriched in hydrocarbons, but the H/D ratio in hydrocarbons will increase as the carrbon number increases For example, let us assume that prior to termination a growing chain has equal amount of H and D atoms and the termiantion requires to add two more H or D atoms with an inverse isotope effect of 0.8. For $C_2$ alkane,the H/D = [2+2*0.44]/[2+2*0.56] = 0.923 while for $C_8$ alkane the H/D = [8+2*0.44]/[8+2*0.56] = 0.974. As shown in Table 4, the H/D ratios from $C_8$ to $C_{14}$ are all less than 1, indicating that there are deuterium enrichment in all of those alkanes. In addition, the H/D ratio decreases as the carbon number increasing, indicating that the deuterium enrichments is a function of carbon number. These results are consistant with above explanation that there is an inverse isotope effect in each step of the propagations.

The deuterium enrichment as a function of carbon number can be explained by the alkylidene mechanism for FT reactions. Assuming that the $C_1$ growing chains during $H_2/D_2$ competition experiemnts are formed statistically, the amounts of the three posible $C_1$ growing chains are: M = $CH_2$, 0.25; M = CHD, 0.5; and M = $CD_2$, 0.25. As shown in Scheme 2A, when $C_1$ growing chain M = $CH_2$ reacts with either monomers MCH or MCD, the fraction to form $MCH_2CH$ = M or $MCH_2CD$ = M is $f_h$ since the same $C_1$ growing chain undergoes hybridization change from $sp^2$ to $sp^3$. We further assume that the rates of formation of $C_2$ growing chains from $MCH_2CH$ = M or $MCH_2CD$ = M through reacting with the MH or MD are the same, which is 0.5. Thus, the amount of each of 4 posible $C_2$ growing chains is $o.25f_h \times 0.5 = 0.5^3f_h$. For producing $C_2$ growing chain from $C_1$ growing chain M = CHD as shown in Scheme 2B, the amount of each of 4 posible $C_2$ growing chains is $0.5f_d \times 0.5 = 2 \times 0.5^3f_d$. Similarly, the amount of each of 4 posible $C_2$ growing chains produced from $C_1$ growing chain M = $CD_2$ (Scheme 2C) is $0.5^3f_D$.

Among the 12 $C_2$ growing chains produced by $C_1$ growing chains, there are 8 distinct $C_2$ growing chains. Their structure and amount are listed in Table 5.

The termination of each of possible $C_2$ growing chains through




**Scheme 1.** Alkylidene Mechanism for Fe catalyzed Fischer-Tropsch Synthesis (A) CO/$H_2$ as the syngas and (B) CO/$D_2$ as the Syngas.

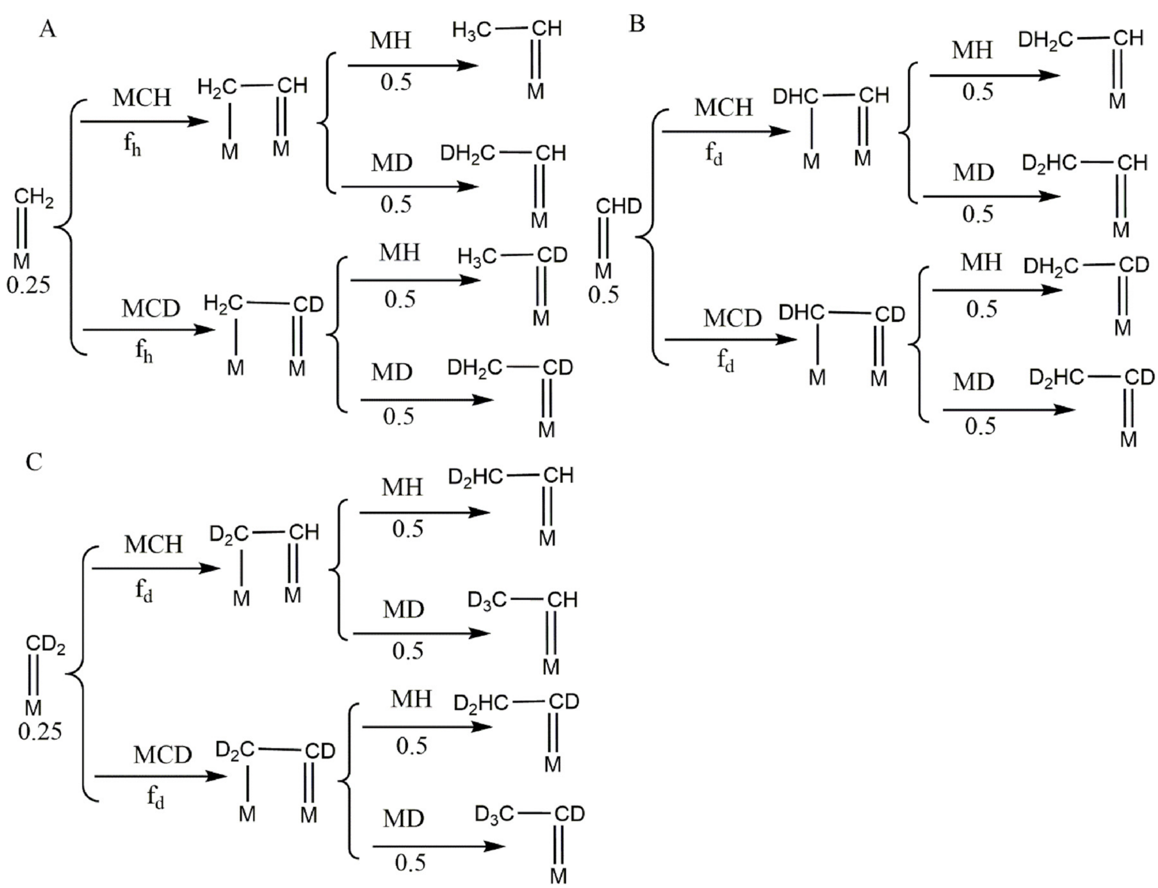
**Table 4**
Ratio of H/D in hydrocarbons produced from $H_2/D_2$ competition experiments.

| Carbon # | Run 1 (H/D) | Run 2 (H/D) | Run 3 (H/D) |
|---|---|---|---|
| 8 | 0.9407 | 0.8819 | 0.9307 |
| 9 | 0.8938 | 0.8470 | 0.8956 |
| 10 | 0.8931 | 0.8495 | 0.8792 |
| 11 | 0.8893 | 0.8438 | 0.8805 |
| 12 | 0.8877 | 0.8379 | 0.8814 |
| 13 | 0.8877 | 0.8421 | 0.8800 |
| 14 | 0.8786 | 0.8481 | 0.8769 |

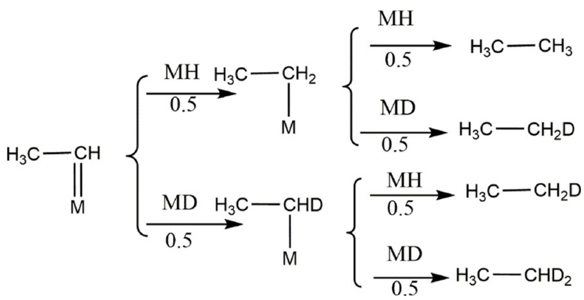
reacting with either MH or MD will produce 4 corresponding ethane isotopemers as shown in Scheme 3 by using the termination of $CH_3CH = M$ as the example. Again, we assume that there is no isotope effect during termination step when the growing chain reacting with

**Table 5**
Calculated amount of $C_2$ growing chains after one step of propagation assuming a 1:1 mixture of $H_2$ and $D_2$.

| Chemical Structure | Amount (x $0.5^3$) |
|---|---|
| $CH_3CH = M$ | $f_h$ |
| $CH_3CD = M$ | $f_h$ |
| $CH_2DCH = M$ | $f_h + 2f_d$ |
| $CH_2DCD = M$ | $f_h + 2f_d$ |
| $CHD_2CH = M$ | $3f_d$ |
| $CHD_2CD = M$ | $3f_d$ |
| $CD_3CH = M$ | $f_d$ |
| $CD_3CD = M$ | $f_d$ |




**Scheme 2.** Formation of $C_2$ growing chains (A) from $C_1$ growing chain $M = CH_2$, (B) from $C_1$ growing chain M = CHD and (C) from $C_1$ growing chain $M = CD_2$.




**Scheme 3.** Termination of a $C_2$ growing chan $CH_3CH = M$.

**Table 6**
Theoretical Amount of $C_2$ Alkane Isotopemers Produced by Fischer-Tropsch reactions using equal amount of $D_2$ and $H_2$ Gasses.

| M.F | # of D Atoms | Amount of isotopomers X ($0.5^5$) |
|---|---|---|
| $C_2H_6$ | 0 | $f_h$ |
| $C_2H_5D$ | 1 | $4f_h + 2f_d$ |
| $C_2H_4D_2$ | 2 | $6f_h + 9f_d$ |
| $C_2H_3D_3$ | 3 | $4f_h + 16f_d$ |
| $C_2H_2D_4$ | 4 | $f_h + 14f_d$ |
| $C_2HD_5$ | 5 | $6f_d$ |
| $C_2D_6$ | 6 | $f_d$ |

either MH or MD. Four isotopemers of ethane (ethane-$d_0$, ethane-$d_1$, ethane-$d_1$ and ethane-$d_2$) are formed, and amount of each isotopemer is $0.5^3 f_h \times 0.5 \times 0.5 = 0.5^5 f_h$. Thus, total of 32 possible isotopemers are formed from these 8 $C_2$ growing chains. The amounts of the 7 distinct isotopemers (from ethane-$d_0$ to ethane-$d_6$) of ethane are calculated as shown in Table 6.

The propagation of the 8 $C_2$ growing chains in Table 5 will form 12 distinct $C_3$ growing chains, which in turn will result in 9 isotopemers of $C_3$ alkanes (from propane-$d_0$ to propane-$d_8$) after the termination steps by reacting with MH or MD. The amount of each isotopermer is listed in Table 7.

Following the same procedure, the amounts of isotopemers of $C_4$ alkane, $C_5$ alkane and $C_6$ alkane are presented in Tables 8–10 respectively.

By using the same procedrure, we developed equations for calculating the H/D ratio in each alkane with 2–14 carbon atoms. If there is no inverse isotope effect (IIE) in a FT reaction, then $f_h = f_d = 0.5$, and H/D = 1 for each hydrocarbon. However, if there is an inverse isotope effect, IIE = 0.8 for example, then $f_h = 0.44$, $f_d = 0.56$, the calculated H/D ratio for ethane is 0.9655, and the H/D ratio in tetradecane will be 0.9085 as shown in Table 11, indicating that there exists the deuterium enrichment. In addition, the calculations predict that the H/D ratios decrease with increasing carbon number, which was experimentaly observed during Fe catalazed FT reactions reported in this study as well as in Co catalyzed FT reactions reported previously [11].

As indicated in Table 11, the Alkylidene mechanism predicts that

**Table 7**
Theoretical Amount of $C_3$ Alkane Isotopomers Produced by Fischer-Tropsch reactions using equal amount of $D_2$ and $H_2$ Gasses.

| M.F | # of D Atoms | Amount of isotopomers X ($0.5^6$) |
|---|---|---|
| $C_3H_8$ | 0 | $f_h^2$ |
| $C_3H_7D$ | 1 | $5f_h^2 + 3f_hf_d$ |
| $C_3H_6D_2$ | 2 | $10f_h^2 + 16f_hf_d + 2f_d^2$ |
| $C_3H_5D_3$ | 3 | $10f_h^2 + 35f_hf_d + 11f_d^2$ |
| $C_3H_4D_4$ | 4 | $5f_h^2 + 40f_hf_d + 25f_d^2$ |
| $C_3H_3D_5$ | 5 | $f_h^2 + 25f_hf_d + 30f_d^2$ |
| $C_3H_2D_6$ | 6 | $8f_hf_d + 20f_d^2$ |
| $C_3HD_7$ | 7 | $f_hf_d + 7f_d^2$ |
| $C_3D_8$ | 8 | $f_d^2$ |

**Table 8**
Theoretical Amount of $C_4$ Alkane Isotopomers Produced by Fischer-Tropsch reactions using equal amount of $D_2$ and $H_2$.

| M.F | # of D atoms | Amount of isotopomers X ($0.5^7$) |
|---|---|---|
| $C_4H_{10}$ | 0 | $f_h^3$ |
| $C_4H_9D$ | 1 | $6f_h^3 + 4f_h^2f_d$ |
| $C_4H_8D_2$ | 2 | $15f_h^3 + 25f_h^2f_d + 5f_hf_d^2$ |
| $C_4H_7D_3$ | 3 | $20f_h^3 + 66f_h^2f_d + 32f_hf_d^2 + 2f_d^3$ |
| $C_4H_6D_4$ | 4 | $15f_h^3 + 95f_h^2f_d + 87f_hf_d^2 + 13f_d^3$ |
| $C_4H_5D_5$ | 5 | $6f_h^3 + 80f_h^2f_d + 130f_hf_d^2 + 36f_d^3$ |
| $C_4H_4D_6$ | 6 | $f_h^3 + 39f_h^2f_d + 115f_hf_d^2 + 55f_d^3$ |
| $C_4H_3D_7$ | 7 | $10f_h^2f_d + 60f_hf_d^2 + 50f_d^3$ |
| $C_4H_2D_8$ | 8 | $f_h^2f_d + 17f_hf_d^2 + 27f_d^3$ |
| $C_4HD_9$ | 9 | $2f_hf_d^2 + 8f_d^3$ |
| $C_4D_{10}$ | 10 | $f_d^3$ |

**Table 9**
Theoretical Amount of $C_5$ Alkane Isotopomers Produced by Fischer-Tropsch reactions using equal amount of $D_2$ and $H_2$.

| M.F | # of D atoms | Amount of isotopomersX ($0.5^8$) |
|---|---|---|
| $C_5H_{12}$ | 0 | $f_h^4$ |
| $C_5H_{11}D$ | 1 | $7f_h^4 + 5f_h^3f_d$ |
| $C_5H_{10}D_2$ | 2 | $21f_h^4 + 36f_h^3f_d + 9f_h^2f_d^2$ |
| $C_5H_9D_3$ | 3 | $35f_h^4 + 112f_h^3f_d + 66f_h^2f_d^2 + 7f_d^4$ |
| $C_5H_8D_4$ | 4 | $35f_h^4 + 196f_h^3f_d + 210f_h^2f_d^2 + 52f_hf_d^3 + 2f_d^4$ |
| $C_5H_7D_5$ | 5 | $21f_h^4 + 210f_h^3f_d + 378f_h^2f_d^2 + 168f_hf_d^3 + 15f_d^4$ |
| $C_5H_6D_6$ | 6 | $7f_h^4 + 140f_h^3f_d + 420f_h^2f_d^2 + 308f_hf_d^3 + 49f_d^4$ |
| $C_5H_5D_7$ | 7 | $f_h^4 + 56f_h^3f_d + 294f_h^2f_d^2 + 350f_hf_d^3 + 91f_d^4$ |
| $C_5H_4D_8$ | 8 | $12f_h^3f_d + 126f_h^2f_d^2 + 252f_hf_d^3 + 105f_d^4$ |
| $C_5H_3D_9$ | 9 | $f_h^3f_d + 30f_h^2f_d^2 + 112f_hf_d^3 + 77f_d^4$ |
| $C_5H_2D_{10}$ | 10 | $3f_h^2f_d^2 + 28f_hf_d^3 + 35f_d^4$ |
| $C_5HD_{11}$ | 11 | $3f_hf_d^3 + 9f_d^4$ |
| $C_5D_{12}$ | 12 | $f_d^4$ |

**Table 10**
Theoretical Amount of $C_6$ Alkane Isotopomers Produced by Fischer-Tropsch reactions using equal amount of $D_2$ and $H_2$.

| M.F | # of D atoms | Amount of isotopermers ($x0.5^9$) |
|---|---|---|
| $C_6H_{14}$ | 0 | $f_h^5$ |
| $C_6H_{13}D$ | 1 | $8f_h^5 + 6f_h^4f_d$ |
| $C_6H_{12}D_2$ | 2 | $28f_h^5 + 49f_h^4f_d + 14f_h^3f_d^2$ |
| $C_6H_{11}D_3$ | 3 | $56f_h^5 + 176f_h^4f_d + 116f_h^3f_d^2 + 16f_h^2f_d^3$ |
| $C_6H_{10}D_4$ | 4 | $70f_h^5 + 364f_h^4f_d + 424f_h^3f_d^2 + 134f_h^2f_d^3 + 9f_hf_d^4$ |
| $C_6H_9D_5$ | 5 | $56f_h^5 + 476f_h^4f_d + 896f_h^3f_d^2 + 496f_h^2f_d^3 + 76f_hf_d^4 + 2f_d^5$ |
| $C_6H_8D_6$ | 6 | $28f_h^5 + 406f_h^4f_d + 1204f_h^3f_d^2 + 1064f_h^2f_d^3 + 284f_hf_d^4 + 17f_d^5$ |
| $C_6H_7D_7$ | 7 | $8f_h^5 + 224f_h^4f_d + 1064f_h^3f_d^2 + 1456f_h^2f_d^3 + 616f_hf_d^4 + 64f_d^5$ |
| $C_6H_6D_8$ | 8 | $f_h^5 + 76f_h^4f_d + 616f_h^3f_d^2 + 1316f_h^2f_d^3 + 854f_hf_d^4 + 140f_d^5$ |
| $C_6H_5D_9$ | 9 | $14f_h^4f_d + 224f_h^3f_d^2 + 784f_h^2f_d^3 + 784f_hf_d^4 + 196f_d^5$ |
| $C_6H_4D_{10}$ | 10 | $f_h^4f_d + 46f_h^3f_d^2 + 296f_h^2f_d^3 + 476f_hf_d^4 + 182f_d^5$ |
| $C_6H_3D_{11}$ | 11 | $4f_h^3f_d^2 + 64f_h^2f_d^3 + 184f_hf_d^4 + 112f_d^5$ |
| $C_6H_2D_{12}$ | 12 | $6f_h^2f_d^3 + 41f_hf_d^4 + 44f_d^5$ |
| $C_6H_1D_{13}$ | 13 | $4f_hf_d^4 + 10f_d^5$ |
| $C_6D_{14}$ | 14 | $f_d^5$ |

the deuterium enrichment is a function of carbon number. As the carbon number of the alkane increaes, more deuterium will be enriched to the alkane but the increasing rate is decreasing as the carbon number increases. As shown in Fig. 2, most of H/D values in alkanes in Fe/Si/K catalysed FT reactions (run 1–3) are very close to the predicted values for the inverse isotope effects being between 0.67 and 0.75. For run 1 and 3, the H/D ratios for all alkanes except $C_8$ are very close to the ones predicted by the mechanism with IIE = 0.75. For run 2, the H/D ratios in all of alkanes except $C_8$ are also very close to the predicted with

**Table 11**
Calculated H/D ratios in hydrocarbons with different inverse isotope effect (IIE).

| Carbon # | IIE = 0.68 | IIE = 0.70 | IIE = 0.75 | IIE = 0.80 | IIE = 0.90 |
|---|---|---|---|---|---|
| 2 | 0.9437 | 0.9473 | 0.9565 | 0.9655 | 0.9831 |
| 3 | 0.9129 | 0.9188 | 0.9333 | 0.9473 | 0.9744 |
| 4 | 0.8949 | 0.9020 | 0.9196 | 0.9366 | 0.9692 |
| 5 | 0.8830 | 0.8910 | 0.9105 | 0.9295 | 0.9657 |
| 6 | 0.8747 | 0.8832 | 0.9041 | 0.9244 | 0.9633 |
| 7 | 0.8685 | 0.8774 | 0.8994 | 0.9207 | 0.9615 |
| 8 | 0.8637 | 0.8730 | 0.8957 | 0.9178 | 0.9600 |
| 9 | 0.8598 | 0.8694 | 0.8927 | 0.9154 | 0.9589 |
| 10 | 0.8567 | 0.8665 | 0.8903 | 0.9135 | 0.9580 |
| 11 | 0.8541 | 0.8640 | 0.8883 | 0.9120 | 0.9572 |
| 12 | 0.8519 | 0.8620 | 0.8867 | 0.9106 | 0.9565 |
| 13 | 0.8501 | 0.8603 | 0.8852 | 0.9095 | 0.9560 |
| 14 | 0.8485 | 0.8589 | 0.8840 | 0.9085 | 0.9555 |

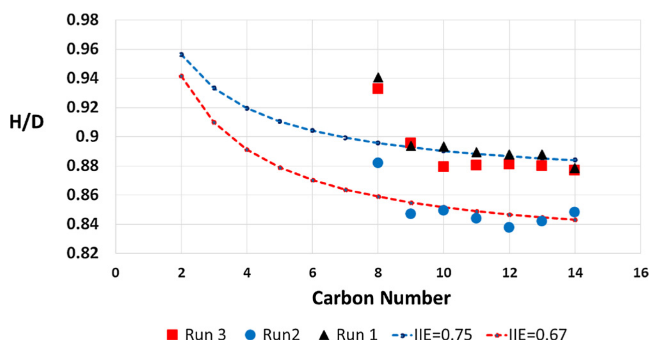



**Fig. 2.** The H/D ratios in alkanes observed during Fe/Si/K catalysed FT reactions and predicted by Alkylidene mechanism.

IIE = 0.67. In Fe catalyzed FT reactions, significant amount of alkanes with carbon number less than 10 could be produced from the readsorption of conresponding alkenes. These alkanes could have less deuterium because 5 of the H (D) atoms in the alkene can undergo H/D exchange [6]. The $^{14}$C study showed that the amount of readsorption of alkenes with carbon number more than 10 in a Co catalyzed FT reaction is very small [18]. Probably that is why H/D ratio of $C_8$ alkane is larger (less dueterium) than predicted.

## 4. Conclusions

There is an inverse isotope effect during FT reactions. This phenominon was reported by several research groups over the years using different techniques [11,15,16]. The technique used in this study is $H_2$/$D_2$ swiching method, which requires mesearing the CO conversion and $CO_2$ formation. Based on many years' studies by many groups, it is safe to conclude that there is inverse isotope effect during FT reactions.

Deuterium enrichment was observed in abiogenic hydrocarbons that was believed to be produced through a FT-like process in Earth crust [19]. By using a $H_2/D_2$ competition method, we observed the phenominon of deuterium enrichment in hydrocarbons produced by Co catalyzed FT reactions [11], and in this study we observed the similar results in Fe catalyzed FT reactions. We also observed that the deuterium enrichment is a function of carbon number. The inverse isotope effect and the deuterium enrichment in hydrocarbons are experimental facts of the FT reactions. Any feasible mechanism for FT reactions should be able to explain these two experimental facts as well as other facts.

The alkylidene mechanism for FT reaction has been used to explain the inverse isotope effect by which the growing chain is RCH = M and the monomer is M≡CH, and the inverse isotope effect originats from the hybridization changing in RCH = M from sp$^2$ to sp$^3$ in the new growing chain $RCH_2CH$ = M during C—C bond coupling [11,12]. This mechanism suggests that the invese isopope effect occures in every step of the propagation steps of the FT reaction, resulted in the values of $[CO_{conv}]^H/[CO_{conv}]^D$ or $[HC]^H/[HC]^D$ or $[C_1{}^+]^H/[C_1{}^+]^D$ much smaller than expected for an inverse isotope effect originated from hybridization changes from sp$^2$ to sp$^3$ during C—C bond formation. Based on the alkylidene mechanism and ASF equation, the inverse isotope effect of Fe catalized FT reactions is calculated to be $0.7-0.95$.

The alkylidene mechanism can explain the deuterium enrichment in hydrocarbons produced by FT reactions. Since the inverse isotope effect occurs in every step of propagation, more deuterium containing compounds will be produced and the deuterium will be enriched more as the carbon number increases in a reaction that equal amount of $H_2$ and $D_2$ is used. This study showed that the H/D ratio in hydrocarbons produced by a FT reaction can be calculated based on the mechanism if the isotope effect of the reaction is known. In the meantimes, if there exists the deuterium enrichment in a FT reaction, the inverse isotope effect can be estimated by the distribution of H/D ratios in hydrocarbons.

The deuterium enrichment in hydrocarbons can be used as an indicator to determine whether the hydrocarbons are produced through FT-like reactions [19]. According to alkylidene mechanism, deuterium will not be enriched in methane since its formation from $C_1$ growing chain requires C–H bond formation only. Therefore, to determine the origin of hydrocarbons, analyzing the deuterium content in ethane or $C_2{}^+$ hydrocarbons should be able to provide more valuable information.

Fischer-Tropsch synthesis is a very complex reaction. The intend of this study is to provide an alternative explanation for the inverse isotope effect reported over the years by many reaserch groups. Even though the Alkylidene mechanism is suported by $H_2/D_2$ experiments, a lot of questions regarding this mechanism remain to be answered. For example, we still do not know the exact structures of the growing chain RCH = M. If we have a clear picture for the growing chain, it would help us to understand the formation of $C_1$ species, the termiantion steps and the FT reactions.

## CRediT authorship contribution statement

**Buchang Shi:** Conceptualization, Investigation, Formal analysis, Writing - original draft, Writing - review & editing. **Yunxin Liao:** Investigation, Methodology, Formal analysis, Writing - original draft. **Zachary J. Callihan:** Investigation. **Brad T. Shoopman:** Investigation. **Mingsheng Luo:** Formal analysis.

## Acknowledgements

We thank the URC of Eastern Kentucky University and NSF-EPSCoR through contract number EPS-0814194 for financial support.

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